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JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY 9744/04

Paper 4 Practical **15 August 2017**

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **19** printed pages.

[Turn over

Answer **all** questions.

1 During the light dependent stage of photosynthesis, hydrogen ions and electrons are transferred to hydrogen acceptor molecules, including NADP.

DCPIP (2,6-dichlorophenolindophenol) is a blue dye, which acts as a hydrogen ion and electron acceptor. As DCPIP accepts hydrogen ions or electrons it is reduced and becomes colourless.

You are required to investigate the effect of different wavelengths of light on the rate of the light dependent stage of photosynthesis in a leaf extract containing chloroplasts.

You are provided with:

- a leaf extract in buffered solution, labelled **L**, in a beaker with ice,
- DCPIP solution, labelled **D**,
- filters that allow light of specific wavelengths to pass through, as shown in Table 1.1.

Table 1.1

colour	label	wavelength / nm
purple	P	425
blue	B	450
green	G	525
orange	O	625
red	R	675

The leaf extract is an irritant. It is recommended that you wear safety goggles/glasses and gloves.

Proceed as follows.

1. Stir the leaf extract, **L**, using the glass rod.

2. Use a syringe to draw up 0.5 cm^3 of **L**.

3. Wipe the outside of the syringe to remove any liquid.

4. Put the syringe at the centre of a white tile. This will be used as a colour standard.

5. You are required to add enough DCPIP solution, **D**, to change the colour of the remaining leaf extract, **L**. The change in colour must be sufficient to be observable in the 0.5 cm^3 sample transferred to a syringe in step **8**.
   - Using a Pasteur pipette, put about 0.5 cm^3 of DCPIP solution, **D**, into the remaining leaf extract, **L**, in the specimen tube.
   - Shake the specimen tube gently so that the colour spreads evenly.
   - Tilt the specimen tube and view the colour against a white background.
   - If there is no noticeable colour change, add DCPIP solution, **D**, drop by drop until a noticeable colour change is achieved.

6. Immediately wrap the specimen tube containing the mixture of **L** and **D** in foil. Cover the specimen tube with a foil lid, as shown in Fig. 1.1. This should be easy to remove to obtain the mixture of **L** and **D**. Put back the covered specimen tube containing the mixture of **L** and **D** in the beaker with ice.

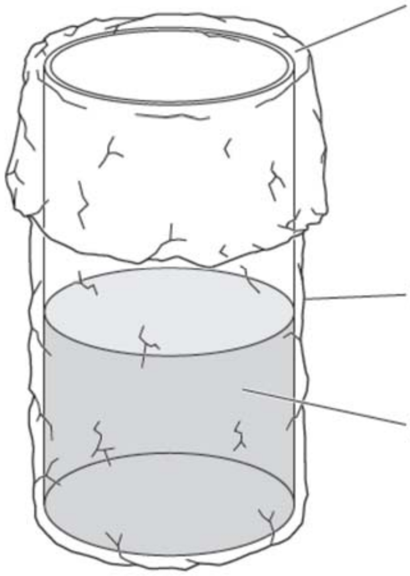


Fig. 1.1

7. Place the bench lamp 10 cm from the syringe on the white tile. Do **not** switch the lamp on.

   The next steps have to be carried out very quickly one after another, so read steps 8–15 and refer to Fig. 1.2 before proceeding.

8. Remove the foil lid and use a clean syringe to draw up 0.5 cm^3 of the mixture of **L** and **D** in the specimen tube. Replace the foil lid immediately.

9. Wipe the outside of the syringe and place it next to the colour standard on the white tile. This is the test syringe.

10. Immediately cover both syringes with the purple filter, **P**, as shown in Fig. 1.2 on page 4.

11. Switch on the bench lamp and immediately start a stopwatch or stop clock.

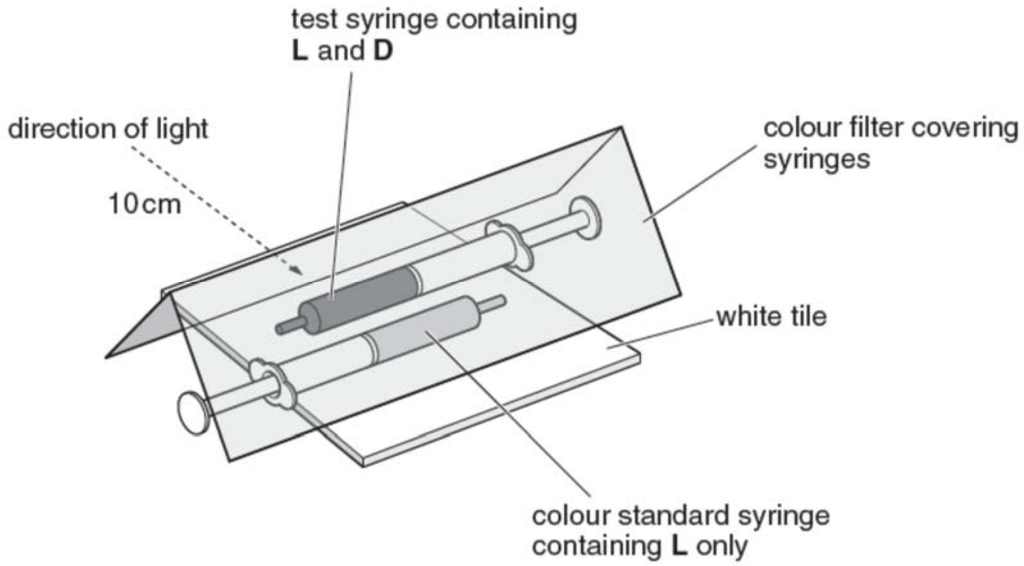


Fig. 1.2

12. In the space provided in **(a)**, record the time taken for the colour in the test syringe to match that of the colour standard. If the colour does not match after 300 seconds then record 'more than 300'.

13. Switch off the bench lamp.

14. Expel the contents of the test syringe into the beaker labelled **waste**. Rinse the syringe.

15. Repeat steps **8–14** using each of the four remaining coloured filters in turn (blue, green, orange and red).

(a) Record these results in a suitable table in the space provided to show the effect of wavelength on the time to decolourise DCPIP. [3]

Table showing the effect of wavelength on the time to decolourise DCPIP

Wavelength / nm	Time to decolourise DCPIP / s
425 (purple)	113
450 (blue)	241
525 (green)	more than 300
625 (orange)	155
675 (red)	142

1. **Suitable column / row headings with correct units;;**
2. **Recording time in whole number of seconds;;**
3. **Record 425 nm (purple) is fastest and 525 nm (green) is slowest or 'more than 300' and results recorded for all five filters;;**

(b) (i) Give one reason to explain why the leaf extract was kept on ice. [1]

1. **To prevent the enzymes in the extract from damaging the chloroplasts;;**

(ii) State why the leaf extract containing DCPIP was kept covered by foil. [1]

1. **To prevent photosynthesis / light dependent reaction occurring and decolourising the DCPIP;;**

(iii) Describe a suitable control that could have been set up for this investigation. [1]

1. **Description of a tube containing DCPIP and water / boiled leaf extract only;;**
2. **Description of a tube containing DCPIP and leaf extract in the dark;;**

(c) Suggest why the rate of photosynthesis is different at different wavelengths. [2]

1. **Some wavelengths are used more effectively than others for photosynthesis or uses data to make the same point, e.g. purple / red / orange wavelengths are the most effective;;**
2. **Chlorophyll absorbs some wavelengths of light more than others or uses the data to make the same point, e.g. chlorophyll absorbs purple / blue wavelengths and orange / red / long wavelengths;;**
3. **Some wavelengths release more H^+ ions / electrons than others, e.g. purple / red / orange wavelengths;;**

(d) Suggest two significant sources of error in this experiment and describe two corresponding improvements that could be made to reduce the effects of these errors. [4]

1. **Difficulty in matching the colour of the standard by eye;;**
2. **Use a colorimeter;;**

3. **Difficulty in observing the colour all the time because of keeping the filter in place;;**
4. **Filter in front of the syringes so colour can be seen from behind;;**

5. **Leakage of light through the ends of the folded filter;;**
6. **Remove all other light sources;;**

7. **There is warming up / increase in temperature as experiment proceeds due to the heating effect of the lamp;;**
8. **Use of a heat filter or cool light source;;**

In another similar investigation, a student collected leaves from two varieties of the same species of a garden plant that has different coloured leaves

Variety A dark red leaves
Variety B green and white striped leaves

The student made a chloroplast extract from the leaves of each variety and measured the rate of photosynthesis for each extract in different wavelengths of light.

Table 1.2 shows the rates of photosynthesis calculated by the student from her results.

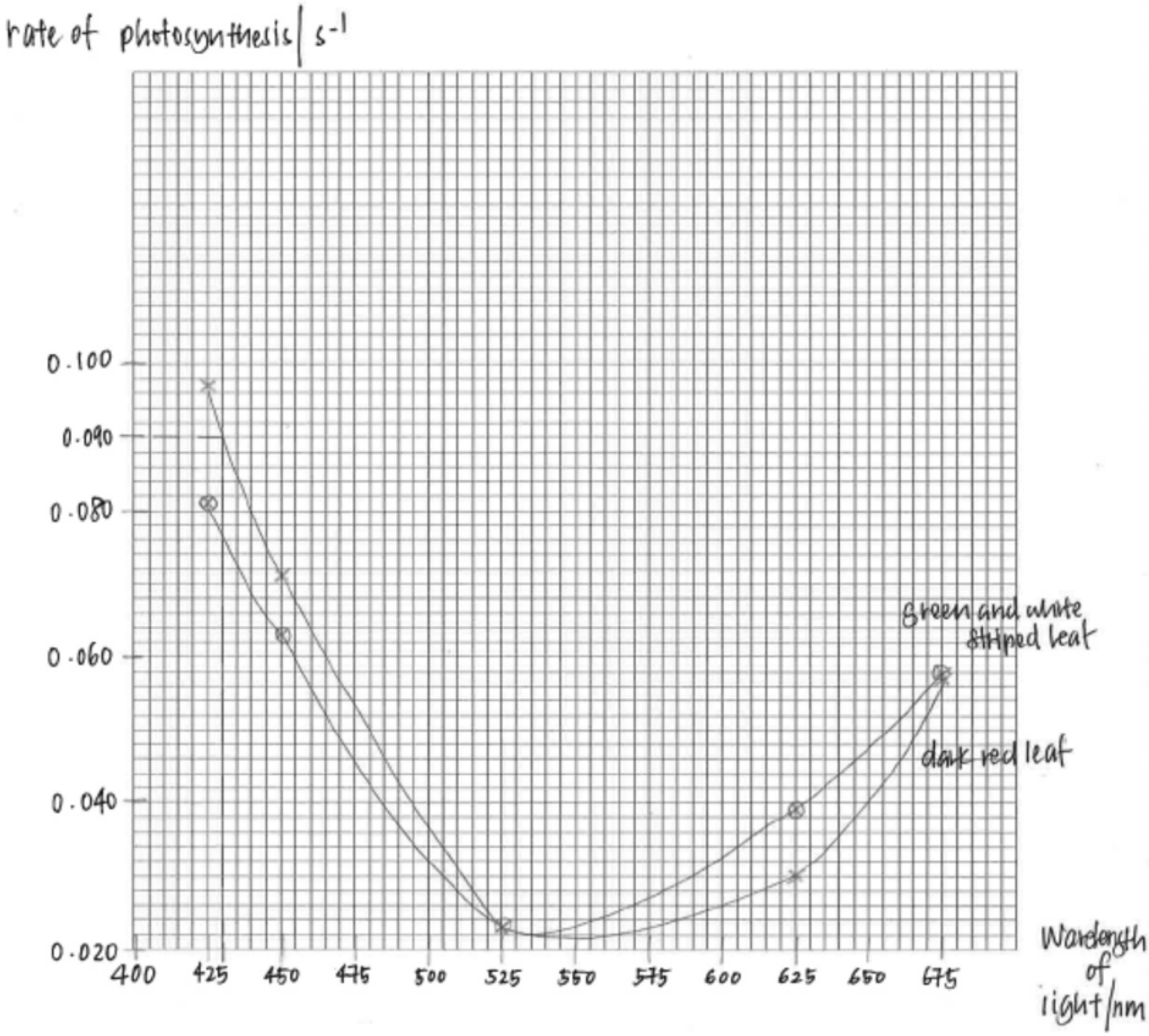
Table 1.2

wavelength of light / nm	rate of photosynthesis / s^{-1}	
	source of chloroplasts	
	dark red leaf	green and white striped leaf
425	0.097	0.081
450	0.071	0.063
525	0.023	0.023
625	0.030	0.039
675	0.057	0.058

(e) Use the grid provided to plot line graphs showing the effect of wavelength on the rate of photosynthesis. [4]

1. **Axes correctly labelled with correct units;;**
2. **Scaled appropriately with ascending scale and equidistant intervals, with a scale such that plotted points occupy at least 50% of the graph paper in both the x and y directions;;**
3. **All values from student's calculations of rate plotted correctly to ± half a small square on the graph paper provided;;**
4. **Line graph showing best-fit line + 2 clearly labelled graph drawn;;**

Graph of rate of photosynthesis s^{-1} against wavelength of light / nm



(f) Use you graph to estimate the rate of photosynthesis at a wavelength of 450 nm in plants with dark red leaves. [1]

1. 0.090 s^{-1};;

(g) Explain how light of wavelength 450 nm leads to the decolourisation of DCPIP. [1]

1. Light energy causes <u>release of electrons from chlorophyll / from photolysis of water</u> and <u>electrons reduce the DCPIP</u>;;

(h) The photosynthetic pigments of the leaves from the two varieties of plants were extracted and were separated by two-way chromatography. The pigments were first separated by one solvent and then separated again by a second solvent at right angles to the first solvent. Fig. 1.3. shows the results for the two different varieties.

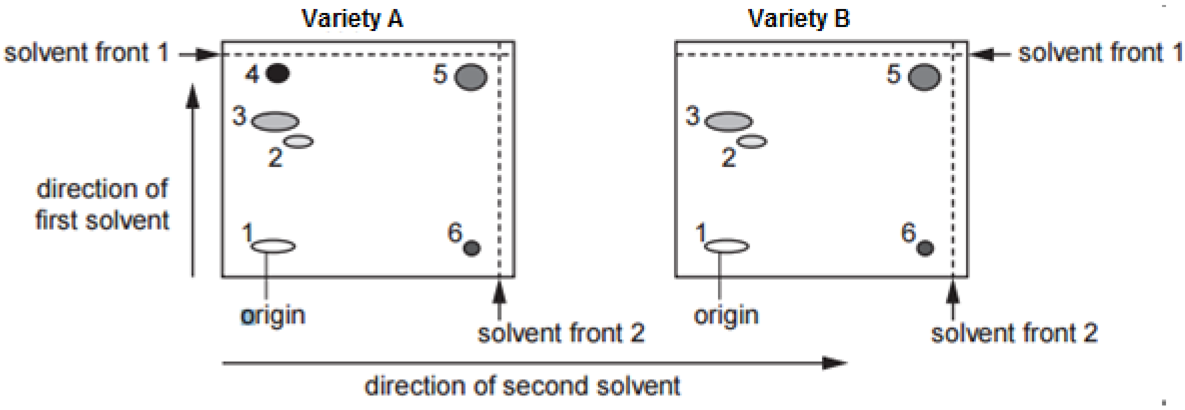


Fig. 1.3

Different photosynthetic pigments absorb different wavelengths of light. Table 1.3 shows some information about the pigments, P, Q, R, S and T, found in the 2 varieties, including the wavelength of light at which maximum light absorption occurs.

Table 1.3

pigment	wavelength of light / nm	Rf value	
		solvent 1	solvent 2
P	620	0.20	0.89
Q	545 and 547	0.60	0.29
R	420 and 660	0.65	0.11
S	490	0.91	0.19
T	430 and 645	0.82	0.92

$$Rf = \frac{\text{distance moved by pigment}}{\text{distance moved by solvent front}}$$

One of the varieties lacks one of the pigments.
Using the information in Table 1.3 and Fig. 1.3:

(i) identify the variety that lacks one of these pigments and state the letter of the missing pigment. [1]

1. Variety B and pigment S;;

(ii) state the evidence that supports your answer to (i). [2]

1. Chromatogram for Variety B has a pigment / spot / number 4 missing;
2. At about Rf 0.91 (in solvent 1) / Rf 0.19 (in solvent 2) / it has the highest Rf in solvent 1 / a low Rf in solvent 2;;

[Total: 22]

2 Methylene blue stains dead cells blue.
Living cells are not stained blue so they will appear white or clear.

You are provided with:
- methylene blue solution, M, (handle carefully as it will stain your skin)
- suspensions of yeast cells, labelled S1, S2 and S3.

Each suspension, S1, S2 and S3 has been heated for ten minutes at 45°C or 80°C or 100°C.

You are required to:
- use the microscope to observe the colour of the yeast cells from S1, S2 and S3, after M has been added
- record your observations by using annotated drawings of three yeast cells from each of S1, S2 and S3
- identify the temperature at which each of S1, S2 and S3 was heated.

1. Label three microscope slides S1, S2 and S3.

2. Place one drop of S1 onto slide S1 and add one drop of M. Mix carefully using a glass rod. (If M comes into contact with your skin rinse with cold water.)

3. Repeat step 2 with S2 and S3.

4. Leave for five minutes.

5. Add a coverslip to each slide.

6. Use the paper towel to dry off any excess liquid around the coverslip.

7. Use the microscope to observe the yeast cells on each slide, then select cells which you can draw and annotate to describe the effect of the methylene blue, M.

(a) (i) Prepare the space below and record your observations by:
- making drawings of three cells from each of the slides in the boxes provided
- annotating your drawings to describe the effect of methylene blue, M on the cells. [4]

S1

S2

S3

1. **At least 9 separate cells in total drawn in boxes S1, S2 and S3 + size at least <u>10mm across smallest cell</u>, in any box;;**
2. **<u>Drawing and quality</u>: Do not give mark for any shading, any ruled lines, or any line is too thick, has any feathery or dashed lines or gap in line, has any overlaps;;**
3. **Drawn only <u>3 cells</u> in each of the three boxes;;**
4. **At least one colour stated for each of the cells in the boxes S1, S2 and S3;;**

(ii) Use your observations to identify the temperature that was used to heat each of the suspensions S1, S2 and S3.
Complete the table. [1]

suspension	temperature / °C
S1	**100**
S2	**45**
S3	**80**

(iii) Explain how you identified the yeast cells that had been heated at 100 °C. [1]

1. **Yeast cells blue + therefore inactive or dead;;**

(b) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, and the stage micrometer, find the actual length, in μm, of one of the yeast cells that you have drawn in S2.

Show the measurements that you made and your working. [3]

1. **shows division of stage micrometer measurement by number of eyepiece graticule divisions;;**
2. **shows measurement of yeast cell from S2 in eyepiece graticule divisions;;**
3. **(conversion of measurement from S2, in eyepiece graticule divisions, to) correct answer to calculation in μm;;**

<u>x40 objective</u>

<u>20</u> graticule units	**=**	**<u>10</u> micrometer divisions**
	=	**<u>10</u> x 0.01 mm**
	=	**<u>0.10</u> mm = 100 μm**
So 1 graticule unit	**=**	**<u>5.00</u> μm (3 sf);;**

Length of a yeast cell = ________ of graticule units;;

Length of a yeast cell = ________ μm;;

Actual length of a yeast cell = …………………………... μm

(c) Draw a straight line on your drawing across the yeast cell to show where you took your measurement. [1]

Use your knowledge of the actual size of the yeast cell to calculate the magnification of your drawing. [1]

1. **Magnification of drawing = Length of the drawing / actual length of guard cells**

= ________ ;;

(d) The yeast *Rhodotorula glutinis* produces an enzyme, α-arabinofuranosidase, that could be used in the production of compounds to enhance the flavour and smell of fruit juices. The effect of the initial pH of the culture medium on the growth rate of this yeast was tested. Three continuous culture systems were set up, each with a different initial pH. The cultures were sampled at hourly intervals for 20 hours at each pH. The mean growth rate was then calculated.

The mean growth rates with their standard deviations are shown in Table 2.1.

Table 2.1

pH	mean growth rate / arbitrary units h^{-1}
4.0	0.156 $\pm$ 0.001
5.2	0.197 $\pm$ 0.013
7.0	0.037 $\pm$ 0.011

A t-test was carried out on the results for pH 4.0 and pH 5.2 and gave the value,

$$t = 2.4$$

The degree of freedom is 38.

Based on the findings of the *t*-test, a student concluded that pH 5.2 was optimum for the production of the enzyme α-arabinofuranosidase by *R. glutinis*. Suggest two reasons why this conclusion may not be valid. [2]

1. **Only three pH values tested / only 2 pH values (used for *t*-test);;**
2. **No data between pH 4 and 5.2 / 5.2 and 7;;**
3. **Only growth measured;;**
4. **Yield of enzyme might be higher at different pH than optimum growth;;**

(e) A student carried out *t*-test on the results to compare the lengths of yeast cells when grown in different media.

A number of *t*-test was carried out to find out if, after 70 minutes, the difference in mean yeast cell length is significant:

1. between medium A and medium B $t = 2.50$
2. between medium A and medium C $t = 3.56$
3. between medium B and medium C $t = 1.94$

Table 2.2 shows the critical values for the *t*-test.

The number of degrees of freedom is 18.

Table 2.2

degrees of freedom	10	12	14	16	18	20	22	24	26	28	30	40	50	60
probability 0.05	2.23	2.18	2.14	2.12	2.10	2.09	2.07	2.06	2.06	2.05	2.04	2.02	2.01	2.00
probability 0.01	3.17	3.06	2.98	2.92	2.88	2.85	2.82	2.80	2.78	2.76	2.75	2.70	2.68	2.66

State what conclusions can be drawn about the significance of the differences in mean lengths from the three values of t given above. [3]

1. **Critical value (at p >0.05) is 2.10;;**
2. **A + B (1) and A + C (2) have values greater than the critical value / 2.10 / p<0.05**
   OR
   B + C (3) has value less than critical value / 2.10 / p>0.05;;
3. **A + B / A + C results are significant / not due to chance / caused by an environmental factor**
   OR
   B + C results are not significant / due to chance;;
4. **(A + B / A + C) Differences due to chance is less than 1 in 20 / 0.05 probability;;**
5. **A + C (also) significant at >p = 0.01;;**

(f) Fig. 2.1 is a photomicrograph of a stained transverse section through part of a plant leaf. This plant species is native to part of Asia.

You are not expected to have studied this leaf.

Fig. 2.1

Draw a large plan diagram of the part of the leaf shown in Fig. 2.1. On your diagram, use a ruled label line and label to show the vascular bundle. [4]

1. **At least 2 lines for upper epidermis and 2 lines for lower epidermis + one enclosed area + size at least 80mm for depth of midrib + no shading;;**
2. **No cells + one enclosed area (vascular bundle);;**
3. **Correct proportion of vascular bundle in relation to distribution of tissues in midrib;;**
4. **Uses label line and label to vascular bundle;;**

Low power / Plan drawing of transverse section through part of a plant leaf

[Total: 20]

3 A number of plant tissues are coloured because the cells contain chemicals called betacyanins.

You are provided with beetroot which contains betacyanins which coloured the beetroot red.

In this experiment, you will test the effect of two different alcohols – methanol and ethanol on beetroot membranes. Ethanol is found in alcoholic beverages. Methanol, sometimes referred to as wood alcohol, can cause blindness and death.

If beet membranes are damaged, the red pigment will leak out into the surrounding environment. The intensity of color in the environment should be proportional to the amount of cellular damage sustained by the beet.

Plan an investigation to find out whether or not betacyanin leakage for beetroot occurs at the same intensity using ethanol and methanol.

You must use:

- beetroot
- 40% ethanol
- 40% methanol
- colourimeter

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc.
- stopwatch
- distilled water
- white tile
- scalpel
- forceps

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

(a) Independent and dependent variables

1. **State and describe independent variable;;**
2. **State and describe dependent variable;;**

- Independent variable is concentration of alcohols – 0%, 10%, 20%, 30%, 40% ethanol and methanol.

- Dependent variable is absorbance.

(b) Controlled/fixed/standardised, variables to improve accuracy or reliability

3. **Identifies at least two variables to be controlled;;**
4. **Describes how two identified variables are controlled;;**

- Size / shape / number of beetroot pieces used.
  - Use a fixed number of pieces for each experiment.
- Source / age of beetroot pieces
  - All beetroot pieces must come from the same beetroot.
- Volume of alcohol used
  - 10 cm^3 used for each experiment.
- Duration
  - 5 min for each experiment.
- pH
  - Use a constant volume of pH buffer for all experiments.
- Temperature
  - Use a constant temperature of 35°C by placing test-tubes in a thermostatically controlled water bath.

(c) Scientific reasoning

5. **Describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible;;**

- Reference to phospholipids / proteins of membrane being affected + Explanation of how membrane component is affected e.g. movement or solubility of phospholipids / denaturation of proteins leading to effect on the permeability of the membrane (cell or vacuole);;
- Suitable trend suggested for stated factor e.g. Increasing alcohol concentration increase membrane permeability + Result in more betacyanins / pigment / red colour leaking from the cells / vacuoles;;
- Membrane permeability can be measured by measuring the extent of leakage of pigment using a colourimeter to give an absorbance value;;
  (Max 2)

(d) Describe the experimental procedure (method)

6. **Plans suitable method to vary the concentration of ethanol and methanol;;**

- Describe simple dilution methods + Table showing how to dilute 40% ethanol and methanol to get 10%, 20%, 30%.

7. **Plans suitable method to get equal amounts of beetroot pieces for ethanol / methanol and experiment;;**
8. **Wash and rinse beetroot in distilled water to remove any pigments which may leak out during cutting;;**

1. Cut beetroot to cubes of 1cm by 1cm by 1cm (or any reasonable fixed sizes).
2. Wash and rinse beetroot in distilled water to remove any pigments which may leak out during cutting.
3. Place 5 beetroot cubes (any reasonable numbers) into a test tube containing 10 cm^3 of 10% ethanol.

9. Specifies how to ensure all beetroot pieces are exposed to the alcohols / are submerged in the alcohols;;

4. Carefully agitate the tube to make sure that the beetroot pieces do not stick together / Ensure that all beetroot pieces are submerged in the alcohols.

10. Plans a <u>suitable procedure that involves monitoring the betacyanins leakage</u> in response to <u>different concentration of alcohols</u> over a <u>set period of time</u>;;
11. Devises a method that uses tubes, beetroot pieces, ethanol and methanol;;
12. Specifies <u>method of monitoring and recording</u> betacyanins leakage;;

5. Place test-tube in a waterbath / thermostatically controlled waterbath maintained at 35-40˚C (to choose only one temperature).
6. Immediately start timing for 5 minutes using a stopwatch.
7. After 5 minutes, transfer 1 cm^3 of solution from test-tube to a cuvette and measure the colour intensity using a colorimeter. Record the absorbance values.

8. Repeat step 1-7 to obtain another 2 readings.
9. Repeat step 1-8 for 20%, 30%, 40% and 0% (control) ethanol.
10. Repeat entire experiment twice.
11. Repeat step 1-10 for the different concentrations of methanol.
12. Plot a graph of absorbance / A.U. against concentration of ethanol and methanol / %.

(e) Draw and annotate relevant diagram(s) – marking points may be credited from diagrams. e.t.c.

(f) Ensuring reliability and accuracy
13. Describe how to ensure reliability (replicates and repeats);;
14. Describe how to ensure accuracy (more or wider range of concentrations used);;

(Reliability)
- Repeat two more times with different / new beetroot pieces and alcohols.

(Accuracy)
- Repeating with more or wider range of alcohol concentrations.

For teacher perusal
- Insufficient number of values of independent variables e.g. only at 0%, 10%, 20%, 30% and 40%, so lack of results between intervals, changes could be missed so making conclusions about trends are less valid.
  Modification: Include intermediate concentrations within the range e.g. 5%,15% ,25%, 35%.
- Insufficient range of independent variables (0-40%), lack of results beyond the range investigated could make it difficult to identify a trend or pattern.
  Modification: extend the range e.g. 0-60%.

(g) Recording

15. Shows how results are to be presented in the form of tables with independent (alcohols concentration) and dependent variables (absorbance) in appropriate columns / rows;;

<u>Table showing absorbance value with increase concentration of ethanol</u>

Concentration of ethanol / %	Absorbance value / A.U.
0	
10	
20	
30	
40	

<u>Table showing absorbance value with increase concentration of methanol</u>

Concentration of methanol / %	Absorbance value / A.U.
0	
10	
20	
30	
40	

(h) Risks/safety

16. Refers to hazards and precaution;;

Risks	Safety Precautions
Thermometer and test tube breakage.	Remove the thermometer / test tube from the water bath immediately after use and place it in a safe place e.g. drawer / test tube rack, to prevent it from rolling onto the floor.
Scalpel are sharp objected that should be handle carefully.	Place the sharp objects away from the main work area after use.
Ethanol and methanol are highly flammable.	Ensure that there is no naked flame nearby.
Methanol can cause blindness when splashed into the eyes.	Wear goggles and wash eyes immediately with cold water when methanol splashed into the eyes.

(i) Control

17. refers to control experiment without alcohols;;

- $10cm^3$ of distilled water to replace $10cm^3$ alcohols.

(j) Finding if pigment leakage occurs at the same intensity using ethanol and methanol

18. Description that if the 2 graphs do not superimposed on each other then pigment leakage do not occur at the same intensity (vice versa);;